

Question:
At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<think>

The cerebral cortex is the outer , regulatory mantle of the brain that subs erves higher-order functions such as perception , cognition , motor planning , and voluntary control . Cortical neurons and their connections âĖŹsyn apses and astro cytic support âĖŹtake the longest time to mature : most cerebral cortices are not even close to full my el ination and functional maturity at birth , with regions such as the pre frontal cortex continuing to develop well into adolescence and early adulthood .

- By contrast , subc ortical and brain stem structures begin operating early :
- Brain stem (respiration , arousal , cranial nerve aut onomic control) functions promptly at birth .
 - Cere bell um (balance , coordination , some early sensorim otor timing) contributes to fetal and neon atal motor behavior .
 - Thalam us and hypothalam us (sensory relay , aut onomic regulation) are present and functional in the newborn .
 - Basal gang lia (motor planning / execution circuitry) show early activity in fetal movements and post natal habituation .
 - limbic system (em otional /a ffective circuits) is active pre natally and in the neon ate .

Although the visual system is immature at birth (with some early photic and vestib ular-driven reflex es) , it is typically more developed than the cortex ; the question asks for the least developed system overall .

Therefore , at birth the least developed part of the brain is the cortex .

Correct answer: G . cortex

</think>

< answer >

G . cortex

</answer>

